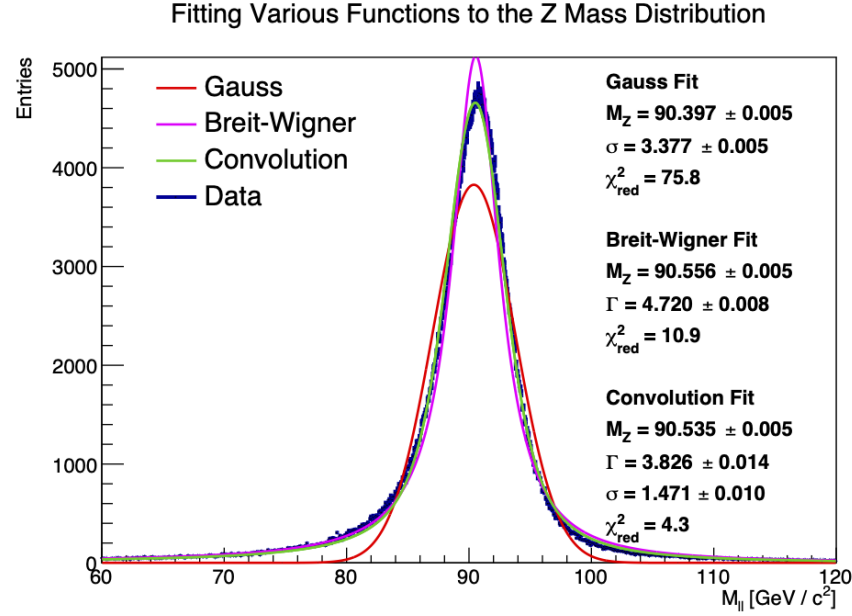


6.6 Fitting the Z Mass

1. Plotting the Fit Functions



2. Extracting Fit Parameters

Using the best fitting model (convolution):

Invariant Mass:

$$M_Z = (90.556 \pm 0.005) \text{ GeV}$$

Decay width (inverse of lifetime):

$$\Gamma = (3.826 \pm 0.014) \text{ GeV}$$

Uncertainty/Standard Deviation:

$$\sigma = (1.471 \pm 0.010) \text{ GeV}$$

Particle Data Group:

$$M_Z = (91.1880 \pm 0.0020) \text{ GeV}$$

$$\Gamma = (2.4955 \pm 0.0023) \text{ GeV}$$

σ -Deviations: